Registration Number:

UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER I) – July, 2019
INTRODUCTION TO STATISTICS - ST 101

Answer **all** questions in the space provided. Time Allowed: **TWO Hours.**

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1. a) Describe the difference between population and sample.

b) To estimate the average GPA of all students at the University of Peradeniya, the average GPA obtained in Mathematics courses were calculated.

i. Identify the variable.

ii. What is the implied population?

iii. What is the sample?

iv. What type of sample was this?

c) If your instructor were to compute the class mean of this examination when it is graded and use it to estimate the average for all tests taken by this class in this semester, would this be an example of descriptive or inferential statistics? Explain.

d) Categorize the following data according to level: nominal, ordinal, interval, or ratio.

i. Length of time to complete an exam. __________

ii. Course evaluation scale: poor, acceptable, good. __________

iii. The name of the city to host the next Olympics. __________

iv. The time of day the marathon starts. __________

v. Egg sizes (Small, Medium, Large, Extra Large, Jumbo) __________

[P.T.O.]

2. a) Describe the relationship between Mean, Median and Mode for the following curves.

i. Symmetrical curve

ii. A negatively skewed curve

iii. A positively skewed curve

b) Consider the following annual income distribution for 600 people in Rs. 1000s.

Monthly income (Rs.1000s)	**Frequency**
0 - 100	9
101 -150	51
151- 200	120
201 - 300	240
301 - 500	136
501 - 750	33
751 - 1000	9
1001 - 1500	2
	$\Sigma f_i = 600$

i. Compute the mean income in Rs.

ii. Find the standard deviation.

iii. Determine the Q_1, Q_2 and Q_3.

[P.T.O.]

2. b) iv. Find the Pearson's Median Skewness.

v. How is the data skewed?

3. a) State the the Chebyshev's Theorem.

b) Heights of 20-year-old males have a bell-shaped distribution with a mean 176.8 cm and a standard deviation of 3.6 cm.

i. About what proportion of all such men are between 173.2 and 180.4 cm tall?

[P.T.O.]

3. b) ii. What interval centred on the mean should contain about 95% of all such men?

c) A population is known to have a mean of 50 and a standard deviation of 6.

(i) Find an interval that contains at least 75% of the data.

(ii) Find an interval that contains at least 24/25 of the data.

(iii) At least what portion of data contained in the interval from 26 to 74?

[P.T.O.]

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4. The following sample data concerns the number of years a student studied French in school versus their score on a proficiency test.

Years	**Test Score**			
3	57			
4	78			
4	72			
2	58			
5	89			
3	63			
4	73			
5	84			
3	75			
2	48			

a) Identify the independent variable and the dependent variable.

Independent variable:
Dependent variable:

b) Draw a scatter plot of the data.

c) Find the equation of the least squares line for this data and write the equation in the form of: $\hat{y}=\beta_0+\beta_1 x$.

[P.T.O.]

4. d) Use the estimated line from (c) to predict the score on the proficiency test of a person who had 3.5 years of French.

e) Compute the correlation coefficient, r_p for this data.

f) What does this coefficient suggest about a linear relationship between the number of years French was studied in school and test scores for this sample?

g) Compute the coefficient of determination and interpret what it means.

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